

L5a: Single Asset Geometric Brownian Motion and NPV

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Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

By the end of this lecture, you will be able to:

- Interpret out-of-sample predictions: explain how historical mean growth and volatility determine the distribution of later prices.
- Calculate an NPV target probability: connect the discounted trade payoff to the probability of exceeding a target at the sale date.
- Update GBM parameter estimates: apply exponentially weighted updates for mean growth and volatility, explain the half-life, and use the estimates in a forecast with fixed parameters.

Lectures and Examples

Example: Out-of-Sample Single-Asset GBM
How well do historical estimates describe later prices? Simulate 2025 prices, construct prediction bands, and compare observed coverage across assets.

Example: GBM Net Present Value Trade Rule
How likely is a trade to exceed a target scaled NPV? Calculate the probability, check it at the median, and examine how it changes with the target.

Example: Updating GBM Parameters with EMA
Can recent observations improve the forecasts? Update mean growth and volatility, then compare the resulting forecasts with the frozen model.

Company Profile: Renaissance Technologies

Renaissance Technologies (RenTec) was founded by Jim Simons (1938–2024), a mathematician and former chair of mathematics at Stony Brook University.

The firm uses mathematical and statistical methods to guide investments. Its researchers come from mathematics, physics, computer science, and related fields.

Its Medallion Fund, launched in 1988, averaged about 66% annually before fees and 39% after fees over 1988–2018, using the returns reported in Gregory Zuckerman's *The Man Who Solved the Market*. These are arithmetic averages of annual returns, not compound growth rates. The fund is closed to outside investors; RenTec's other funds use different strategies.

Sources: RenTec overview; Zuckerman's book; Bradford Cornell's study of Medallion.

Renaissance Technologies: Explore Further

Jobs and internships. Browse the RenTec careers page for research, programming, and systems roles. No internships were listed as of September 18, 2026; check for new openings. The Simons Foundation also offers paid summer research internships through Summer at Simons at the Flatiron Institute.

Watch Jim Simons discuss his work:

- Numberphile: full-length interview (YouTube channel).
- TED: The mathematician who cracked Wall Street (YouTube channel).

Connection to today: Renaissance builds statistical models of prices and tests them against data. Today we do the same at classroom scale with GBM: fit the model to 2014–2024 prices, forecast 2025, and compare the forecast with what happened. Our examples do not describe RenTec's proprietary strategies.

Concept Review: Single Asset GBM Model

Recall that geometric Brownian motion (GBM) describes share price changes through drift and random fluctuations, both proportional to the current price:

$$\frac{dS(t)}{S(t)} = \underbrace{\mu\, dt}_{\text{drift}} + \underbrace{\sigma\, dW(t)}_{\text{random fluctuations}} .$$

Here, $S(t)$ is the share price (dollars per share), with initial price $S(0) = S_0 > 0$, and $dW(t)$ is a Wiener-process increment.

With time measured in years, the constant parameters are:

- Drift $\mu \in \mathbb{R}$: the expected proportional price change per unit time (year^{-1}).
- Volatility $\sigma > 0$: sets the size of the random fluctuations (year$^{-1/2}$).

Concept Review: Growth and Volatility

Over an observation interval $\Delta t > 0$ years, the growth rate is:

$$g_j = \frac{1}{\Delta t} \ln\left(\frac{S_{t_j}}{S_{t_{j-1}}}\right), \qquad t_j = j\Delta t.$$

Under GBM, its mean and standard deviation are:

$$\underbrace{\mu_g = \mathbb{E}[g_j] = \mu - \frac{\sigma^2}{2}}_{\text{mean growth rate}}, \qquad \underbrace{\sqrt{\mathrm{Var}(g_j)} = \frac{\sigma}{\sqrt{\Delta t}}}_{\text{growth-rate standard deviation}}.$$

Drift μ and mean growth rate μ_g both have units of year^{-1}, but differ by $\sigma^2/2$.

From the sample growth-rate standard deviation σ_g, we estimate the volatility parameter as:

$$\hat{\sigma} = \sigma_g \sqrt{\Delta t}.$$

Concept Review: The One-Step Transition

For a grid of N steps, $t_j = j\Delta t$, the exact transition from L4b is:

$$\boxed{S_{t_j} = S_{t_{j-1}} \exp\Big[\mu_g \Delta t + \sigma\sqrt{\Delta t}\, Z_j\Big], \qquad j = 1, \dots, N.}$$

Here, $Z_1, \dots, Z_N$ are independent standard normal draws, one per step.

For a single horizon $T > 0$, we can sample the terminal price directly:

$$S_T = S_0 \exp\Big[\mu_g T + \sigma\sqrt{T}\, Z\Big], \qquad Z \sim \mathcal{N}(0,1).$$

Repeated one-step draws generate a price path. The fixed-horizon form samples the price at the selected time T.

Concept Review: Estimated Parameters

In L4b, we estimated the mean growth rate and volatility from historical data, giving $\hat{\mu}_g$ and $\hat{\sigma}$, where the hats indicate estimates from data. These two estimates are all we need to simulate prices with the one-step transition.

Why, then, calculate $\hat{\mu}$? The GBM differential equation uses the drift μ. To express the fitted model with that parameter, we calculate:

$$\boxed{\hat{\mu} = \hat{\mu}_g + \frac{\hat{\sigma}^2}{2}.}$$

Substituting this drift estimate into the one-step transition, the half-variance terms cancel:

$$\hat{\mu} - \frac{\hat{\sigma}^2}{2} = \left(\hat{\mu}_g + \frac{\hat{\sigma}^2}{2}\right) - \frac{\hat{\sigma}^2}{2} = \hat{\mu}_g.$$

Both forms give the same simulated prices for the same draws of Z_j.

Example: Out-of-Sample Price Predictions

Example: Test predictions against later observations

How well does a model fitted to historical prices describe prices observed afterward?

- Use the mean growth rate and volatility estimated from 2014–2024 data.
- Simulate prices for 2025 and compare prediction bands with observed prices.
- Measure the fraction of dates with observed prices inside each band, first for one firm and then across the dataset.

Next question: what is the probability that a scheduled sale exceeds a target scaled NPV?

GBM Trade Rule

Consider a scheduled trade: buy $n_0 > 0$ shares at S_0 at time 0 and sell at S_T at time $T = N\Delta t$. Assume no dividends or trading costs.

Let g_y be the constant, continuously compounded benchmark growth rate (year^{-1}) corresponding to yield y. The NPV is given by:

$$\mathrm{NPV}(g_y, T) = \underbrace{-n_0 S_0}_{\text{purchase today}} + \underbrace{n_0 S_T e^{-g_y T}}_{\text{discounted sale}}.$$

Dividing by the initial investment gives the scaled NPV, our discounted fractional return:

$$\rho_T = \frac{\mathrm{NPV}(g_y, T)}{n_0 S_0} = \left(\frac{S_T}{S_0}\right) e^{-g_y T} - 1.$$

A positive ρ_T means the present value of the sale exceeds the purchase cost, and a negative ρ_T means it falls short. A zero ρ_T means the trade earns exactly the benchmark growth rate g_y.

Substitute the GBM Solution

Using the mean growth rate $\mu_g = \mu - \sigma^2/2$, substitute the GBM price into the scaled NPV:

$$\rho_T = \exp\left[\underbrace{(\mu_g - g_y)}_{\text{excess growth rate}}\; T + \sigma\sqrt{T}\, Z\right] - 1, \qquad Z \sim \mathcal{N}(0,1).$$

Discounting replaces μ_g with the excess growth rate $\mu_g - g_y$ relative to the benchmark, while leaving the random shock unchanged. The quantity $1 + \rho_T$ is lognormal, so:

$$\ln(1+\rho_T) \sim \mathcal{N}\big((\mu_g - g_y)T, \sigma^2 T\big)\,.$$

A lognormal variable is always positive, so $\rho_T > -1$: the most we can lose is the purchase price $n_0 S_0$.

When $|g_y|T \ll 1$, discounting is small and $\rho_T \approx S_T/S_0 - 1$. We keep the exact discounted expression so the calculation also applies to longer holding periods.

Terminal Target Probability

Choose a target scaled NPV $\rho_\star > -1$. Adding one and taking the logarithm preserves the strict target condition:

$$\rho_T > \rho_\star \quad \Longleftrightarrow \quad \ln(1+\rho_T) > \ln(1+\rho_\star).$$

Substituting the GBM expression gives:

$$(\mu_g - g_y)T + \sigma\sqrt{T}\, Z > \ln(1+\rho_\star).$$

Subtracting $(\mu_g - g_y)T$ and dividing by $\sigma\sqrt{T} > 0$ gives a threshold for the standard normal random variable Z:

$$Z > z_\star, \qquad \boxed{z_\star = \frac{\ln(1+\rho_\star) - (\mu_g - g_y)T}{\sigma\sqrt{T}}.}$$

Thus $Z > z_\star$ means the trade exceeds its target at the scheduled sale. Selling when a boundary is first reached requires a separate calculation.

The Closed-Form Target Probability

The standard normal cumulative distribution function $\Phi(z_\star)$ gives the probability that $Z \leq z_\star$. Subtracting from one gives the probability that $Z > z_\star$:

$$\mathbb{P}(\rho_T > \rho_\star) = 1 - \Phi(z_\star) = 1 - \Phi\left(\frac{\ln(1+\rho_\star) - (\mu_g - g_y)T}{\sigma\sqrt{T}}\right).$$

- For a chosen target, we evaluate $z_\star$ from estimated mean growth and volatility, the holding period, and the benchmark growth rate.
- The formula applies for $T > 0$, $\sigma > 0$, and $\rho_\star > -1$.
- A target of -1 or below is exceeded with probability one because the scaled NPV always exceeds -1 under GBM.

Example: Calculate and verify the NPV target probability

Target Sale Price and the Benchmark

The target scaled NPV corresponds to a required sale price:

$$K = S_0(1 + \rho_\star)e^{g_y T}.$$

With $\rho_\star = 0$, the trade exceeds its target when the sale price beats $S_0 e^{g_y T}$, the value the purchase price would have reached growing at the benchmark rate.

- $g_y = 0$: did the price increase?
- Positive g_y: did it increase enough over the holding period?
- Increasing g_y raises the sale threshold and lowers the probability of exceeding it.

The same prices can be assessed against different assumed growth benchmarks. Choose the benchmark before examining the outcome. We will meet this form again when K is the strike price of an options contract.

Example: GBM Net Present Value Trade Rule

Example: Evaluate an NPV-based GBM trade rule

How likely is a stock trade to exceed our target scaled NPV at a scheduled sale?

We calculate this probability using the selected firm's mean growth and volatility, the holding period, and the benchmark growth rate.

We check the calculation at the median scaled NPV, which must be exceeded with probability one half, then vary the target to examine the probability curve.

Updating GBM Parameters with an EMA

Fixed parameters are a common criticism of GBM. Instead, we can update the estimates as new prices arrive. An exponential moving average (EMA) gives recent observations more weight. For positive prices spaced $\Delta t > 0$ years apart, use the growth rate $g_k = \ln(S_k/S_{k-1})/\Delta t$.

At entry index s, initialize from the training estimates:

$$m_s = \widehat{\mu}_{g,0}, \qquad v_s = \frac{\widehat{\sigma}_0^2}{\Delta t}.$$

Choose a decay factor $0 < \lambda < 1$. For each new observation $k > s$:

$$\begin{aligned} \delta_k &= g_k - m_{k-1}, && \text{(innovation)} \\ m_k &= m_{k-1} + (1-\lambda)\delta_k, && \text{(mean update)} \\ v_k &= \lambda\big[v_{k-1} + (1-\lambda)\delta_k^2\big]. && \text{(variance update)} \end{aligned}$$

Here, m_k and v_k estimate the mean and variance of the growth rate.

EMA: Recover the GBM Parameters

For the growth rate, the estimated mean m_k has units of year^{-1} and the estimated variance v_k has units of year^{-2}. Convert these moments to GBM parameter estimates:

$$\hat{\mu}_{g,k} = m_k, \qquad \text{mean growth rate,}$$

$$\hat{\sigma}_k = \sqrt{v_k \Delta t}, \qquad \text{volatility,}$$

$$\hat{\mu}_k = \hat{\mu}_{g,k} + \frac{\hat{\sigma}_k^2}{2}, \quad \text{drift.}$$

We estimate the volatility parameter by multiplying the growth-rate standard deviation $\sqrt{v_k}$ by $\sqrt{\Delta t}$. The result has units of year$^{-1/2}$.

Variance update: the recursion accounts for the changing mean; it does not include a finite-sample unbiased correction.

EMA: Interpret the Half-Life

The half-life $H_{\text{half}} > 0$ is the number of observations needed to halve the weight of an older observation:

$$\lambda = 2^{-1/H_{\text{half}}}.$$

With a half-life of 21 observations, $\lambda \approx 0.9675$. The updated mean combines 96.75% of the previous mean with 3.25% of the new growth rate.

A shorter half-life makes the estimates respond faster but also makes them more sensitive to noise. The half-life controls the parameter updates; the holding period sets the forecast horizon. Each forecast holds its parameter estimates fixed.

Derivation: Derive the exponentially weighted moments and GBM parameter updates

Example: Updating the Parameter Estimates

Example: Compare frozen and updated estimates

Can giving recent observations more weight improve our forecasts?

Each day, we update the probability that the original trade will exceed its NPV target at the end of the chosen forward window. We compare frozen parameters with EMA estimates of volatility alone or of both mean growth and volatility.

We evaluate all three forecasts against the same observed outcomes to see whether updating improves predictions.

Optional Advanced Material

The L4b examples develop two related questions:

- Uncertainty in mean growth: why can an estimate remain uncertain even with thousands of observations? Examine the observation period, sampling frequency, and their effects on target probabilities.
- Monte Carlo versus the closed form: how many paths are needed to estimate a probability accurately? Compare simulation error with the analytical result and introduce variance reduction techniques.

Examples: Mean-growth uncertainty; Monte Carlo target probabilities.

Summary

We connected GBM price predictions with trade probabilities and parameter updates.

Key takeaways:

- Out-of-sample predictions: we used historical estimates to describe later prices. Prediction bands apply to individual dates.
- NPV target probabilities: we derived the chance of exceeding a target scaled NPV at a scheduled sale for a chosen benchmark growth rate.
- EMA parameter updates: we updated mean growth and volatility from growth-rate moments. The half-life controls how quickly older observations lose weight; each forecast holds its estimates fixed.

Next: L5b extends the price model to correlated assets and portfolio weights.